

Answer: A

11 Using 1st law of thermodynamics,

$$\Delta U = Q + W$$

For change $S \rightarrow Q \rightarrow P$,

$$\Delta U = -8 + 3 = -5 \text{ J}$$

For change $S \rightarrow R \rightarrow P$,

$$-5 = Q + 2$$

$$Q = -7 \text{ J}$$

Answer: B

12 Q is at the amplitude position, hence it must be momentarily at rest. Since the wave is